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COSTS OF SLOWING DOWN GAS EXPLORATION NATURAL IN TWO SCENARIOS OF ENERGY TRANSITION IN COLOMBIA

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Costs of halting natural gas exploration in two energy transition scenarios in Colombia

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Introduction and key messages

This document expands on the argument made by Benavides, Cabrales and Delgado (2022), an article that argued the importance of natural gas and quantified the costs of dismantling natural gas and coal from electricity generation in Colombia between 2023 and 2024. 2035.

This report presents order-of-magnitude figures on energy density and the high costs of electricity produced from non-conventional sources and batteries, reinforcing the need for a transition at a pace that does not harm the country's economy. It also expands the scope of direct cost calculations associated with gas supply constraints, including major consumption sectors such as residential, gas-fired thermal power generation, and industry (excluding coking and refining). The quantification uses the best available information from public sources.

Numerous policy alternatives exist for the energy transition, but to date only one concrete measure has been proposed regarding natural gas: halting exploration—which implies not adding new reserves in the medium term. This document analyzes the costs of two more specific gas transition scenarios:

- **Scenario** 1. It consists of (i) halting new exploration; (ii) NOT building new import infrastructure; (iii) dismantling and replacing natural gas-fired power generation with wind power generation with the same level of reliability; (iv) promoting the electrification of natural gas consumption in the residential and industrial sectors.

- **Scenario** 2. This consists of (i) halting new exploration; (ii) expanding import infrastructure; (iii) maintaining gas-fired power generation capacity; and (iv) maintaining natural gas consumption in industry and households. In cases (ii) and (iii), the reduction in domestic consumption is achieved with increasing proportions of imported gas.

Table 1 presents the cost differential scenarios with respect to Scenario 5 UPME's 2020-2034 Generation and Transmission Expansion Plan, regarding the change in electricity generation supply.

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Table 1. Cost scenarios for domestic gas reduction and modeling implications

Costs	Scenario 1: Accelerated 'zero carbon' type transition	Scenario 2: Transition with increased natural gas imports
ÿCAPEX	-Investment in wind power generation that provides the same level of firmness as natural gas power generation under water stress conditions + asset costs stranded *assets* -Cost of replacing gas boilers with electric boilers in selected industries -Cost of replacing gas stoves with electric induction stoves in homes	Investment in the expansion of the regasification plant in 2026
ÿOPEX	-Difference between costs of energy balance between gas and electricity in selected industries -Difference between costs of energy between gas and electricity for cooking in homes	-Difference between costs of domestic supplies and imported + new molecule transport costs due to flow reconfiguration (network model)

Source: own elaboration.

As can be seen, in Scenario 1 there are capital investment needs in wind power generation, costs of grounded assets, and replacement of equipment on the demand side, while in Scenario 2 there is a capital investment in natural gas import facilities.

The main empirical decisions of this study are: (i) in Scenario 1, asset disposal is carried out linearly between 2023 and 2030; (ii) a perpetuity of the difference in electricity and gas costs was added from 2030 onwards in Scenario 1 and (iii) due to the reduction of domestic reserves and increasing imports, the price of gas to the end user increases between 2023 and 2030 in Scenario 2.

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Costs of Scenario 1

- The net present value of the cost of switching from gas stoves to electric induction cooktops in all households is **COP$18.27 trillion.** Exchanging gas subsidies for electricity subsidies in the lowest income brackets requires an additional **COP$0.74 billions** per year.

- The net present value of the cost of replacing natural gas-fired power generation with wind power generation with equivalent reliability is **COP$ 85.35 billion.**

- The net present value of the cost of replacing natural gas with electricity in boilers in the food products, beverage manufacturing, textiles, leather goods, wood and paper sectors is **COP$ 10.57 trillion.**

Costs of Scenario 2

- The present value of the cost of increasingly importing to compensate for the brake on the Gas exploration is worth **COP$ 3.14 trillion.**

- The difference in the present values of gas transportation costs to 2030, between the scenario with and without exploration, is approximately **COP$ 2.00 trillion.**

- Without the addition of new gas reserves, the increase in the present value of The cost to supply gas to Colombia until 2030 through increasing imports is equivalent to **COP$ 5.14 trillion.**

- By the end of the decade, the natural gas service tariff (molecule) will experience an additional increase of 28.71% in gas costs and 50.95% in transportation costs, compared to the scenario with new reserves. This could imply **an additional increase of more than 25%** in the final tariff.

Reduction of annual CO2 emissions in Scenario 1

- By changing gas cooking to electric induction cooking in 10.4 million households, an annual reduction of **2.5 million tons of CO2eq would be achieved,** which is equivalent to 0.83% of the national total (UNDP, 2022).

- Replacing gas-fired electricity generation with wind power generation would result in an annual reduction of **2.3 million tons of CO2eq,** equivalent to 0.76% of the national total (UNDP, 2022).

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• Changing gas boilers to electric boilers in the food products, beverage production, textiles, leather goods, wood and paper sectors would result in an annual reduction of **0.8 million tons of CO2eq,** equivalent to 0.27% of the national total (UNDP, 2022).

The central messages of this article are as follows:

• No single energy source dominates the others in terms of cost, service continuity, emissions, and volumetric and area density. Natural gas provides firm energy for electricity generation, dense energy for industry and residential consumption at lower costs than liquid fuels, and emits fewer greenhouse gases per unit of energy consumed.

• The energy crisis in Western Europe, which has occurred as a result of Russia's natural gas cuts, demonstrates the precariousness to which the attempt to electrify the entire energy supply with unconventional energy sources leads.
renewable energy and electrifying all final consumption in the short term. Countries are beginning to understand that they must build energy supply portfolios that balance security, diversification, and the reduction of greenhouse gas emissions.

The costs of either scenario overlap with persistent inflationary trends in energy prices observed internationally. In scenario 1, switching from gas cooking to electric induction cooktops in 10.4 million households, replacing gas-fired power generation with wind power, and replacing gas boilers with electric boilers in some productive sectors would reduce CO2 equivalent emissions by 5.6 million tons annually. This would represent 3.2% of the government's commitment to reducing greenhouse gas (GHG) emissions by 2030, at a cost of COP$114.19 trillion. Scenario 2, in turn, highlights the importance of not neglecting domestic gas production.

natural.

The enormous financial effort with such a low impact on the government's commitment is not sustainable for the Colombian economy. Both scenarios demonstrate that natural gas must play a more significant role in the energy portfolio in an international environment where energy security is regaining prominence, and given the threats of electricity generation shortages due to climate shocks such as El Niño.

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Chapter 1. Natural gas as an indispensable energy source in all timeframes

1.1. The physics of the energy transition

With current technologies, a $1 million investment in large-scale solar panels would produce 40 million kWh over a 30-year period, while a $1 million investment in wind power would produce 55 million kWh over the same period. By comparison, an investment in shale oil extraction equipment would generate 300 million kWh over the same period.

It costs less than USD 1 to store a barrel of oil, or a barrel of oil equivalent, in gas form for two months. In comparison, storing the energy equivalent of a barrel of oil with batteries costs USD 200.

Approximately 3 MW of solar or wind power capacity is required for every 1 MW of thermal generators replaced (intermittency effect), even if the cost per kWh is similar. An investment of USD 200,000 in Tesla-type batteries, which weigh 20,000 lbs, can store the energy equivalent of one barrel of oil. For comparison, a barrel of oil weighs 300 lbs and can be stored in a tank that costs USD 20.

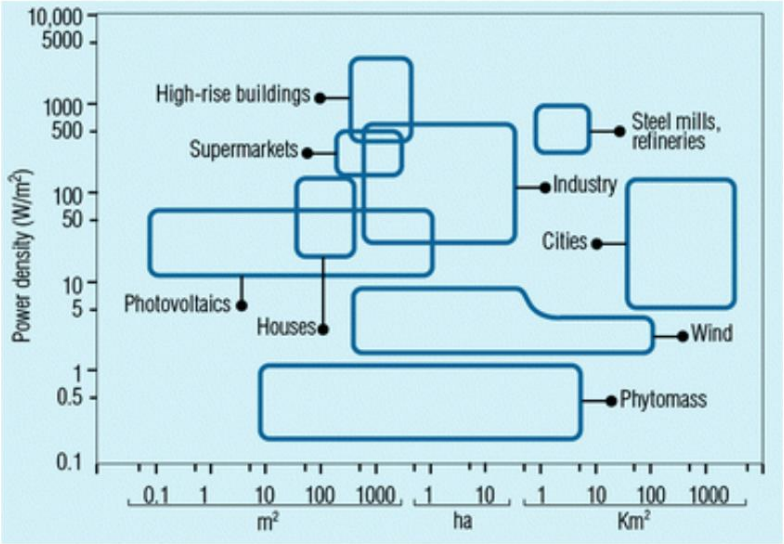
Moore's Law for chips is mistakenly extrapolated to clean energy. Clearly, thermal engines have their limit (Carnot efficiency), but wind farms have the Betz limit (60% capture of kinetic energy from the air, we're currently at 40%); and photovoltaics have the Shockley-Queisser limit (33% of photons are 'converted' into electrons, we're currently at 26%).

Mills (2019) argues that, to completely replace hydrocarbons in the next 20 years, global renewable energy production would need to increase 90-fold. In comparison, it took half a century for oil and gas production to increase tenfold. Renewable energy sources would be required to adopt at a rate nine times faster in almost half the time, with fewer logistical and density advantages.

These characteristics of volumetric density, costs and intermittency, are joined by the very low density per area of the power needed to produce energy through non-conventional renewable energy sources (Smil 2015) and its low alignment with the densities per area of end uses (see Graph 1), and the non-existence of commercial technologies to massively replace the production of steel, concrete, plastic and fertilizers, all linked to hydrocarbons.

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Figure 1. Density misalignment by area between renewable energy supply and major energy demands



Source: Pagliaro et al (2010).

Colombia must expand the use of cleaner energy sources, but, for at least 20 years, not all final energy consumption will be electrified (electrification is at 21%), nor will all electricity supply be produced from renewable sources.

1.2. Why is natural gas required in any supply portfolio of energy?

Per capita energy consumption in Colombia in 2021 was 36 GJ/capita, lower than the Latin American average (50 GJ/capita) and almost half the global average per capita consumption (74.1 GJ/capita). In 2018, the country's energy-related greenhouse gas (GHG) emissions were 1.60 tons of CO2/capita, while those of Sweden, Germany, and Spain were 3.54, 8.55, and 5.52 tons of CO2/capita, respectively. The energy sector is not the main emitter of GHGs; that distinction belongs to the agriculture, forestry, and land-use change sector (accounting for approximately 55% of emissions).

Colombia must increase its per capita energy consumption because it lags behind in investments in the built environment (buildings and infrastructure in the broadest sense), because it has not industrialized, and because energy poverty and vulnerability must be eliminated. A detailed analysis of the possibilities for an energy transition adapted to the context suggests the urgent need to expand supply with a diversified portfolio approach. Natural gas should be vigorously promoted because:

- It is versatile (it can serve as an energy carrier and be consumed in liquid and gaseous states) and has lower GHG emissions per unit of energy than all other fossil fuels. Intermediate and final gas consumption technologies can

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It can replace various more polluting uses without loss of reliability and, in most cases, at a lower cost. It can gradually replace coal and liquid fuels and is a key input for the production of petrochemicals.

- It is a progressively tradable commodity, and its supply chains (production, pipeline, ship, or truck transport), and widely deployed distribution networks, provide dense energy for a variety of end uses. Gas supply and demand technologies have a long engineering tradition and can be designed in a modular fashion.

- It has the capacity to: (i) provide firm electricity generation at low costs and with lower relative emissions than generation with other conventional fuels; (ii) support an aggressive policy of replacing and scrapping older urban transport fleets that consume diesel, with significant co-benefits in particulate matter reduction; (iii) support an energy efficiency policy for direct and indirect thermal uses in district heating systems, with a focus on SMEs, which would also bring the co-benefit of greater competitiveness for consuming firms; (iv) stimulate decentralized energy resource management models through microturbines, to provide local solutions and the sale of surplus energy to the electricity grid; (v) develop high value-added petrochemical products with global demand, such as urea; (vi) help reduce poverty and energy vulnerability; and (vii) contribute to energy continuity and security in industrial and commercial uses that require high volumetric energy density, such as furnaces and boilers.

Measures to orchestrate the presence of natural gas in Colombia's energy mix include, first and foremost, a government commitment to vigorously promote investment throughout the supply chain to ensure its continued availability and its contributions to the country's well-being and growth. Immediately:

- On the supply side, the addition of domestic natural gas reserves should be encouraged under rigorous technical, social and environmental conditions, complemented by imports that are not subject to discretionary supply decisions.

- On the demand side, financing for energy efficiency in end uses must be facilitated; and the entry of Euro 61 gas vehicles into road freight transport must be promoted.

[1] The Euro 6 standard is a regulation established by the European Union (EU) that governs pollutant emissions from new vehicles. It was implemented gradually starting in September 2014.

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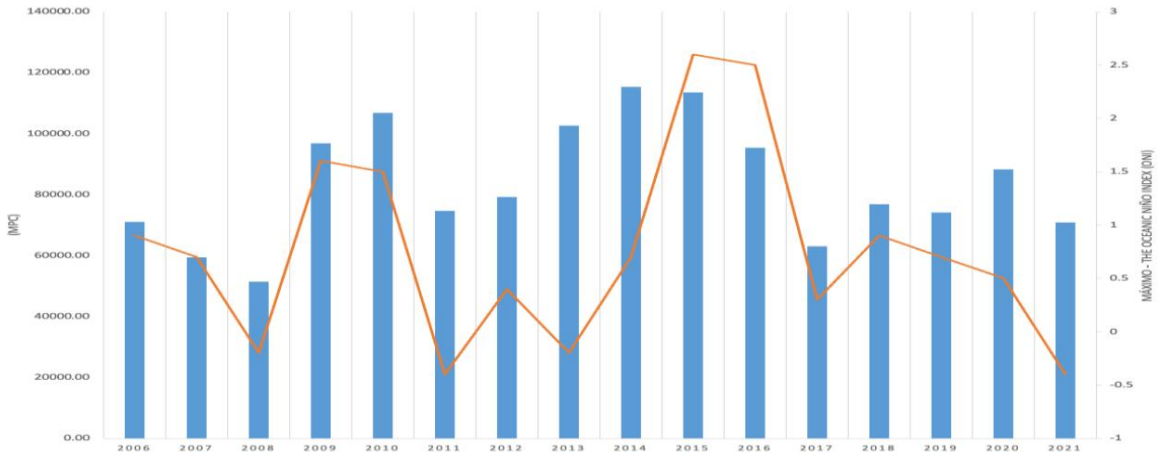
and urban mass transit, which are high-impact niches with high deployment speed.

1.3. What role does natural gas play in the short term in Colombia?

Currently, all Colombians use natural gas directly or indirectly.
Over the past eight years, approximately 10% of the electricity consumed in Colombia has been generated by gas-fired power plants. The number of households using gas for cooking and water heating exceeds 10 million. Furthermore, sectors such as transportation, refining, cement, steel, food processing, paper, and mining also use gas in their production processes, primarily through boilers and furnaces.

Approximately 15% of the gross domestic gas supply, including liquefied natural gas (LNG) imports, is used for thermal power generation. Gas consumption for thermal power plants increases during El Niño events to ensure the reliability of the electrical system, as shown in Figure 2.

Figure 2. Annual consumption of gas-fired thermal power plants versus the maximum of the Oceanic Niño Index (ONI)



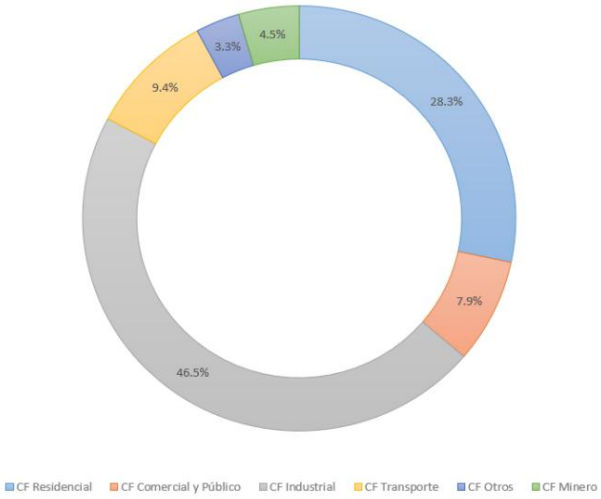
Source: Colombian Energy Balance 2021 (UPME, 2021) and Cold & Warm Episodes by Season (NOOA, 2023)

The supply of gas intended for final consumption is allocated as follows: 46% to the industrial sector, 28% to the residential sector, 9% to the transport sector, 8% to the commercial sector, 4% to the mining sector and 3% to other sectors (Graph 3).

passenger vehicles and in 2015 for light commercial vehicles. The Euro 6 standard sets strict limits for pollutant emissions, mainly nitrogen oxides (NOx) and particulate matter.

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Figure 3. Percentage of final natural gas consumption excluding electricity generation



Source: Colombian energy balance 2021 (UPME, 2021).

1.4. What role will natural gas play in the medium and long term at the international level?

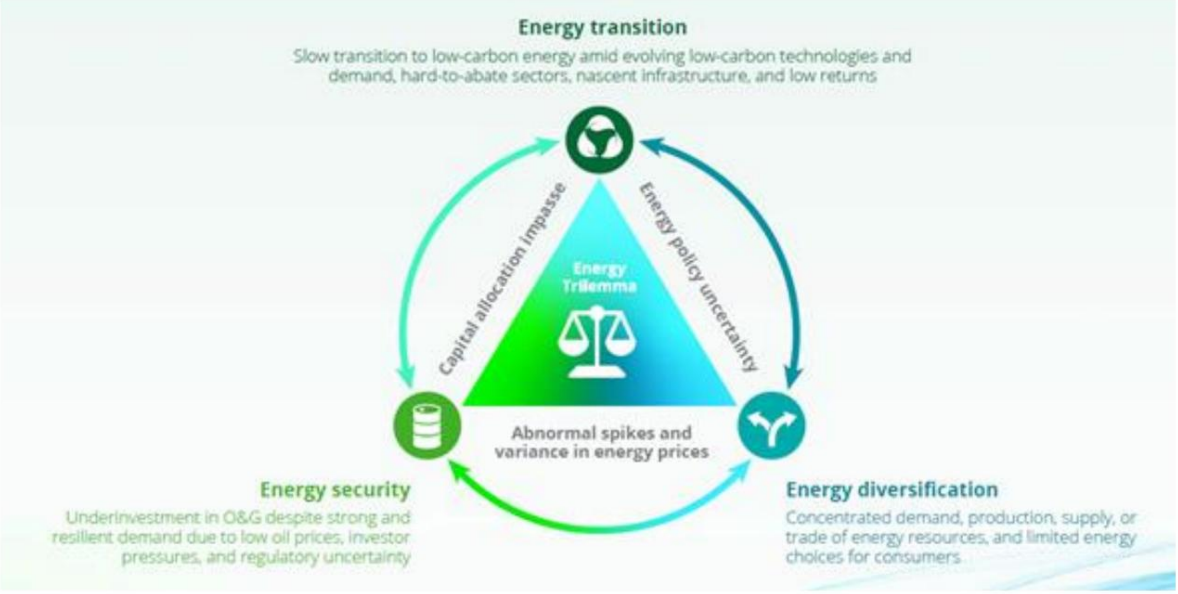
The pandemic and the disruption of value chains, the war between Russia and Ukraine, the threat of cutting off energy exports as a political tool, the vulnerability of energy-importing countries, and the rise in energy and food prices have led to abrupt changes in the energy transition policies of developed countries, moving from the rationing economy to the 'economy of fear' (Peri, 2023).

Some EU countries have included nuclear energy and natural gas within the green energy taxonomy and have invested in LNG import capacity from countries outside Eastern Europe. According to some analysts (Peri, 2023), a period of fierce competition for natural resources, connectivity, cheap labor, and energy can be expected.

This has triggered a readjustment in energy markets, across all fuels and geographies. The components of a balanced energy equation—
Energy security, supply diversification, and the low-carbon transition are under severe pressure or face a trilemma of concerns (Figure 4). Within the energy security pillar, it should be noted that regulatory uncertainty is critical for attracting investments that ensure supply.

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Figure 4. The energy 'trilemma' from 2022



Source: Deloitte (2023).

Although gas prices have reduced their volatility in the summer of 2023, the scar remains of the disruption to energy trade between Europe and Russia that drove global gas markets to new price highs, reaching six to ten times the Henry Hub2 prices .

The energy policies of the United States and Europe began to shift in 2022, following Russia's invasion of Ukraine. As a result, the focus has moved from phasing out natural gas to utilizing it while simultaneously reducing its greenhouse gas (GHG) emissions as cleaner alternatives are developed and deployed. Increased investment in natural gas is expected in 2023, including investments that reduce greenhouse gas emissions. Europe and the United States announced several policies in 2022 to incentivize investment in natural gas and ensure emissions reductions, including:

- EU taxonomy. The EU taxonomy now considers natural gas as an environmentally sustainable economic activity, provided that new natural gas projects replace coal-fired generation, reduce emissions, and are part of a pathway to full conversion to renewable or low-carbon fuels by 2035.

- Joint declaration between the United States and the European Commission on energy security in Europe. The declaration makes commitments to increase the

[2] The Henry Hub index is a price indicator used as a benchmark for natural gas trading in the United States.

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Increase US LNG exports to Europe by 2030 and reduce the emissions intensity of new LNG infrastructure.

- Inflation Reduction Act. This law increases the dynamism of oil and gas lease contracts in the United States. It also imposes fees on excess methane emissions and provides subsidies to oil and natural gas companies to monitor and reduce methane leaks and unproductive combustion.

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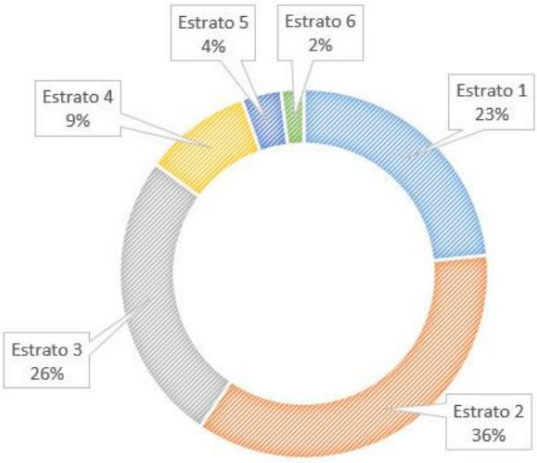
Chapter 2. Methodology for estimating incremental costs of Scenario 1

Scenario 1 consists of (i) halting new exploration; (ii) NOT building new import infrastructure; (iii) dismantling and replacing natural gas-fired power generation with wind power generation with the same level of reliability; (iv) promoting the electrification of natural gas consumption in the residential and industrial sectors.

2.1. Methodology for estimating incremental costs for consumers residential

In Colombia, 10.4 million households use natural gas service. According to the Colombian Energy Balance (BECO) by the UPME (2021), in 2021, 98% of these households were urban and 2% were rural. Figure 5 shows the total number of residential natural gas subscribers by socioeconomic stratum. Households in strata 1 and 2 that use natural gas total 6.2 million, representing 59% of all subscribers.

Chart 5. Percentage by socioeconomic stratum of the number of gas service subscribers (2022)

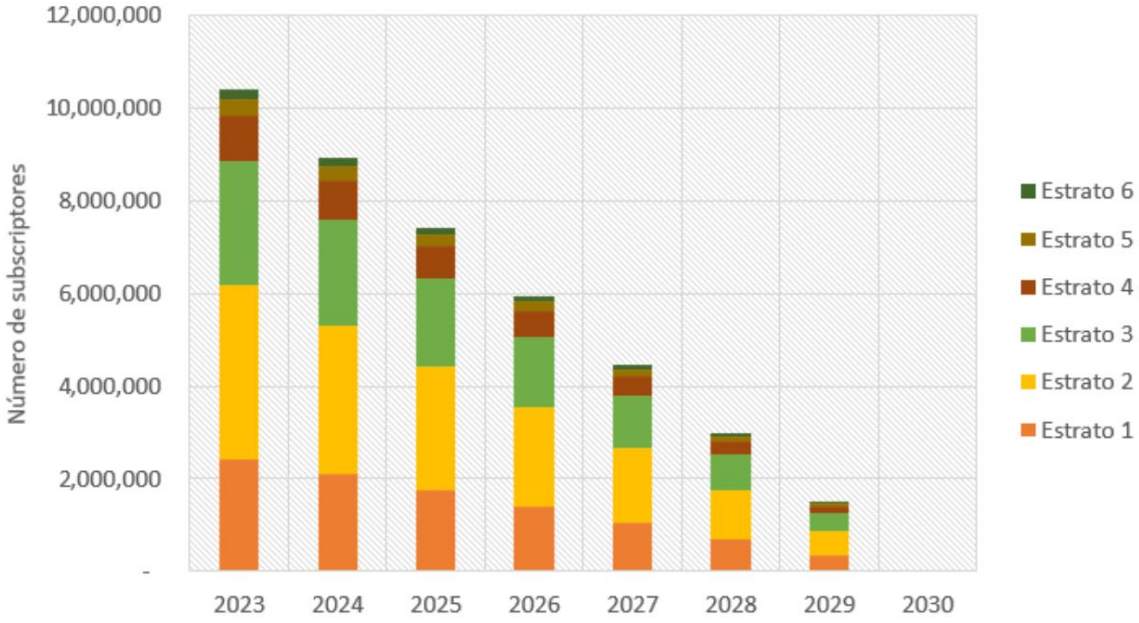


Source: Single Information System for Public Household Services, SUI (2022).

To calculate the difference in costs between using a natural gas stove and an electric induction stove that provides the same level of service in a Colombian home,
The path shown in Graph 6 is followed.

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Figure 6. Path of change to 2030 for residential users of the gas service



Source: own calculations and SUI (2022).

The steps to calculate the difference in generation costs are:

(i) The present value of the cost of natural gas service for cooking is estimated food over a horizon of years:

$$= \ddot{y}_{=1}^{in} \frac{12\ \ddot{y}\ \ddot{y}}{(1+\ )}$$

Where:

^{in} is the cost of gas consumed in stoves over an analysis horizon of years

It is the unit cost ( ) per cubic meter of gas for residential users, which includes the components of supply, transport and distribution.
Users in socioeconomic levels 1, 2, and 3 receive subsidies of 60% and 50%, respectively, of the unit cost. On the other hand, users in socioeconomic levels 5 and 6 pay a 20% contribution on the unit cost.

This is the monthly consumption of a natural gas stove in cubic meters. per household

is the discount rate applied

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(ii) The present value of the cost of electricity service for cooking is estimated
food using induction cookers for a horizon of *T* years:

$$= \ddot{y}_{=1}^{in} \frac{12\ \ddot{y}\ \ddot{y}}{(1+\ )}$$

Where:

^{in} is the total expected value of the electricity consumed in a
analysis horizon of years

This is the unit cost per kWh for residential users, which is the sum of six components: generation, transmission, marketing, distribution, losses, and restrictions. Users in socioeconomic levels 1, 2, and 3 receive subsidies of 60%, 50%, and 15%, respectively, of the unit cost. Users in socioeconomic levels 5 and 6 pay a 20% contribution on the unit cost.

This is the monthly electricity consumption of an induction cooktop
electricity per household and is calculated as the gas consumption for cooking food ( ) multiplied by the energy efficiency factor of a gas stove

(I divide the energy factor of an induction stove:

$$= \ddot{y}\ \frac{\ }{\ _{from}}$$

(iii) The cost of a new electric induction stove is determined ($_{from}$).

(iv) The difference (ÿ) between the present values of gas and electricity service costs and the cost of an induction cooktop is calculated. This represents the present value of the cost to a household of replacing a gas cooktop with an induction cooktop that provides the same level of service for cooking. In Colombia, this difference depends on the household's socioeconomic stratum, which affects gas and energy rates.

$$\ddot{y} = {}^{in}\ {}^{in}\ _{from}$$

Additionally, the reduction in greenhouse gas emissions is quantified.
(ÿ) due to the change of stove, as follows:

$$\ddot{y} = \ddot{y}\ \ddot{y}$$

Where:

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It is the emissions factor of CO2, CH4 and N2O in kilograms of CO2 equivalents per cubic meter of generic natural gas, and

It is the emissions factor of CO2, CH4 and N2O in kilograms of CO2 equivalent per kWh.

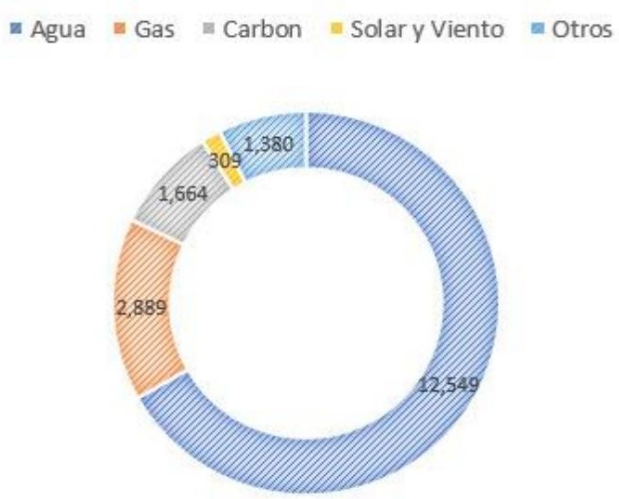
2.2. Methodology for estimating incremental costs in electricity generation

The calculation performed by Benavides, Cabrales and Delgado (2022) is updated.
on a linear path to the complete decommissioning by 2035 of all installed natural gas and coal-fired power generation capacity, and its replacement with wind farms that provide an equivalent level of service. As in the study by Benavides et al. (2022), non-conventional renewable energy generation using solar power is not considered due to its inability to provide power during peak electricity demand in Colombia (7:00 PM).

- 9:00 PM). On this occasion, it is assumed that the gas decommissioning will be carried out gradually until 2030.

Chart 7 presents the existing generation capacity by technology type in the present.

Figure 7. Existing installed generation capacity (2023)



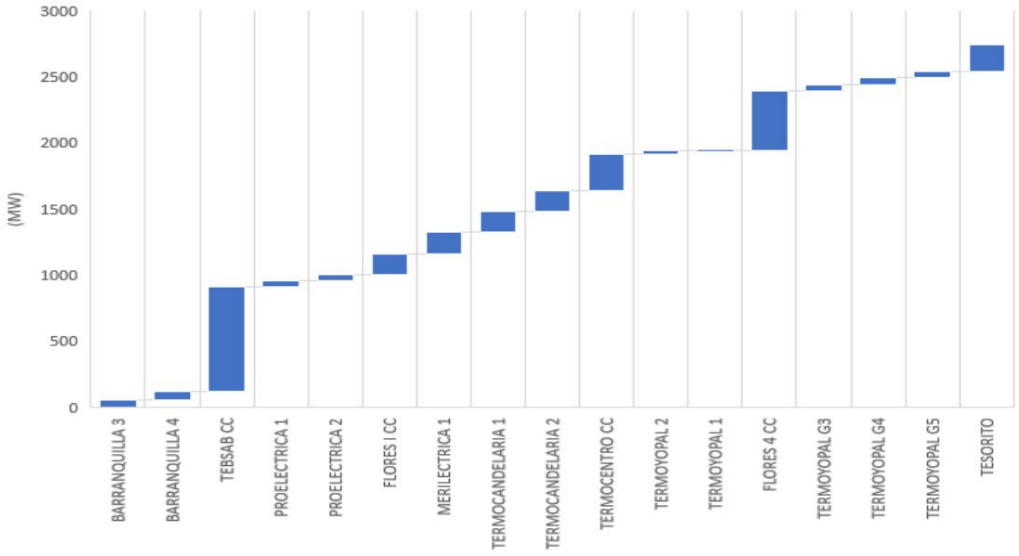
Fuente: XM (2023).

The oldest gas-fired thermal power plants in operation date back to 1980 in Barranquilla, while the most recent are Termoyopal G5 and Tesorito, built in 2022. In total, there are 17 operating gas-fired thermal power plants that contribute significantly and represent 15.4% of the national installed capacity (Figure 8). However, there are also

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There are 14 other gas-fired thermal power plants that are not currently dispatching energy within the wholesale market.

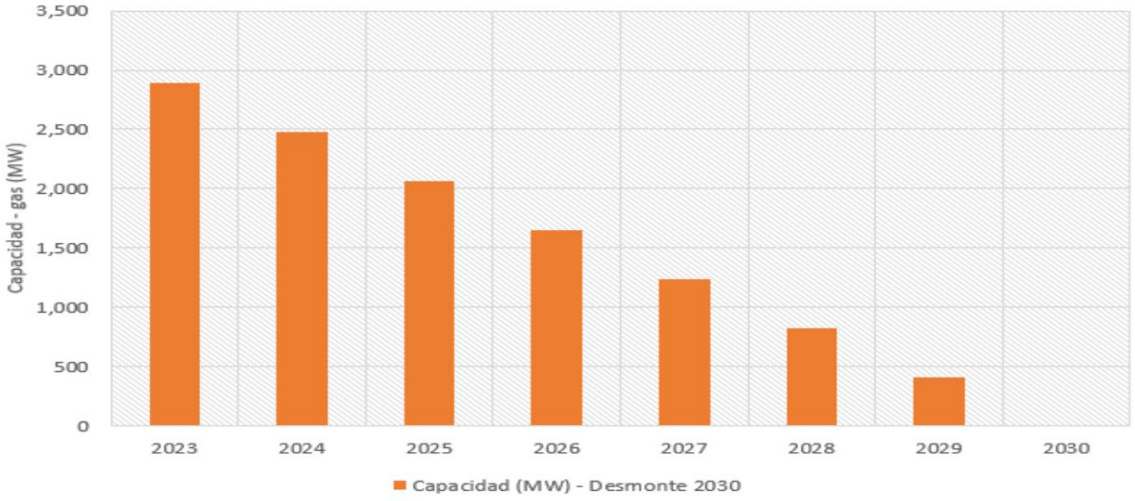
Figure 8. Centrally dispatched gas-fired thermal power plants



Fuente: XM (2023).

To calculate the cost of dismantling gas-fired thermal generation capacity by 2030: (i) Scenario 5 of the UPME's 2020-2034 Generation and Transmission Expansion Plan is used, which is the most likely due to the inclusion of climatic events such as El Niño and La Niña, as well as the entry into operation of the first phase of the Hidroituango project with 1,200 MW; (ii) the removal path of gas-fired generation plants in Graph 9 is followed.

Figure 9. Removal path to 2030 of gas-fired power generation capacity



Source: own calculations and XM (2023).

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The steps to estimate the costs of removing gas-fired thermal power generation are:

(i) The maximum dispatch factor is estimated (ÿ) of gas-fired thermal power plants in order to calculate the effective capacity required with wind power generation:

$$ÿ = \max\{ \frac{ÿ}{} \}$$

Where:

ÿ It is the dispatch of the aggregation of all gas-fired thermal power plants

is the sum of the effective capacities of the plants to gas

(ii) The effective wind capacity required to replace the effective capacity is calculated. installed gas-fired thermal power plants:

$$_{from} = \frac{ÿ \quad ÿ}{ÿ \; _{from}}$$

Where:

ÿ or is the average factor of wind turbines, and

from It is the effective capacity required of wind farms

(iii) The installation cost ( ) required to replace gas-fired generation with the help of the required effective wind capacity is estimated:

$$_{from} = \; _{from} \; _{from} \; ÿ \; _{from}$$

Where:

from from is the total installation cost per kW.

(iv) The cost of replacing gas-fired power generation capacity is calculated using the path shown in Figure 9, based on the effective capacity of wind farms and the cost of installation

(v) The actuarially fair compensation for investors in gas-fired thermal power plants is estimated due to the implementation of a game-changing policy in terms of revenues and costs in regulated sectors, thereby converting their existing investments into stranded assets *by* 2030. The calculation of the value of the stranded assets is based on the net present value of free cash flows

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expected from the gas-fired thermal capacities that would be removed from the system. The projected flows are based on UPME Scenario 5.

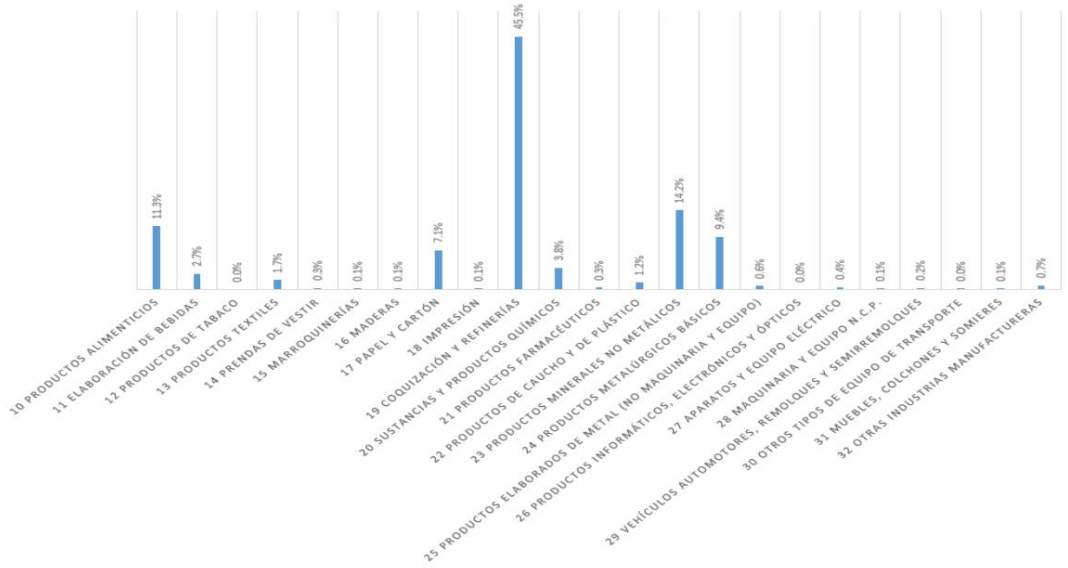
(vi) The cost of replacing existing gas-fired generation with wind generation by 2030 is equal to the cost of replacing existing gas-fired generation with wind generation and the cost of grounded assets.

Additionally, the reduction in greenhouse gas (GHG) emissions generated by dismantling gas-fired thermal power plants is calculated.

2.3. Methodology for estimating the incremental costs of industrial sectors selected

Industrial consumption accounts for 46% of the national total, making it the largest end consumer. According to the Colombian Energy Balance by the UPME (2021), refineries account for 45% of consumption, followed by non-metallurgical mineral products (14%), metallurgical products (9%), food products (11%), and paper and cardboard (7%). Figure 10 shows the percentages of final gas consumption by industries for 2021. The analysis focuses on food products, beverage manufacturing, textiles, leather goods, wood products, and paper, which together account for 20% of total natural gas consumption across all industries. This represents the vast majority of consumption after excluding coking and refining.

Chart 10. Percentage by sector of final gas consumption in industries (2021)

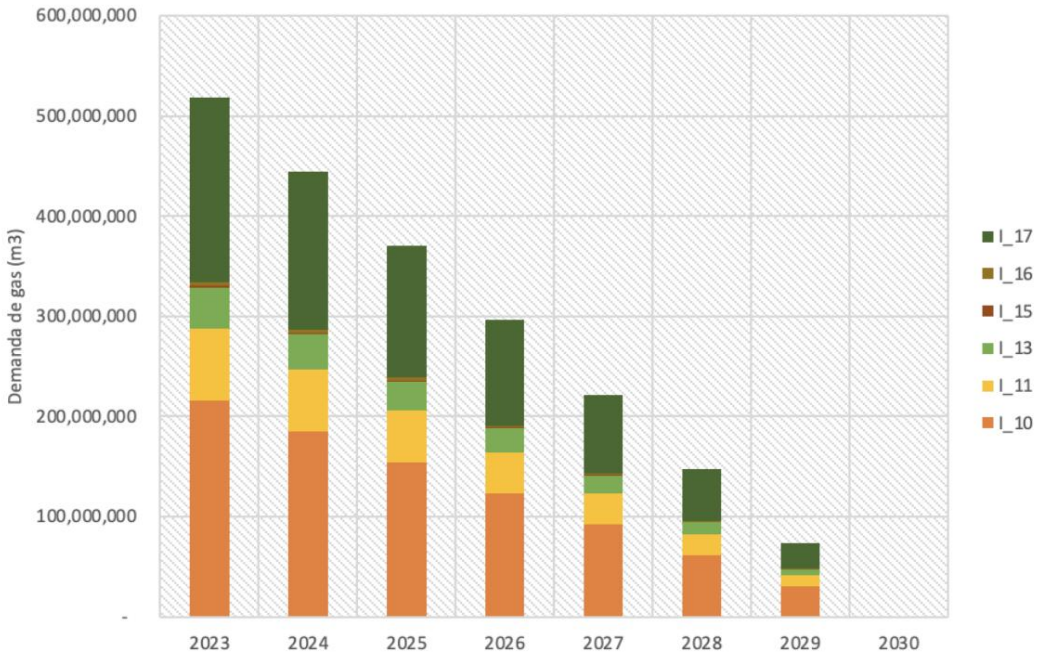


Source: Colombian energy balance, UPME (2021).

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To calculate the direct cost associated with replacing boilers that use natural gas with boilers that use electricity in each industrial sector analyzed, the path of Graph 11 is followed.

Figure 11. Path of change to 2030 from gas boilers to electric boilers



Source: own calculations and BECO (2021).

The steps to estimate costs for the selected industries are:

(i) The present value of the cost of natural gas service for use in boilers is estimated within a timeframe of years:

$$\text{in} = \ddot{y}_{=1} \frac{12\ \ddot{y}\ \ddot{y}}{(1+\ )}$$

Where:

in It is the cost of gas consumed in boilers in years

It is the unit cost (UC) per cubic meter of gas for industrial users, which is the aggregation of the supply, transport, and distribution.

It is the average monthly consumption per sector

is the discount rate applied

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(ii) The annual cost of the electricity service used by the electric boilers is estimated based on the efficiency factors of the gas and electric boilers, and

then its present value, within a timeframe of years:

$$= \ddot{y} \quad \text{---------}$$

$$= \ddot{y} \quad \frac{\ddot{y}}{(1 + \quad)}$$

Where

It is the efficiency of conventional gas boilers

It is the efficiency of electric boilers

is the total expected value of the gas over an analysis horizon of 10 years

The unit cost per kWh for residential users is the sum of six components: generation, transmission, marketing, distribution, losses, and restrictions. Commercial and industrial users pay a 20% contribution on the unit cost.

It is the average monthly consumption per sector

(iii) The cost of a new electric boiler is determined ( )

(iv) The difference (ÿ) between the present costs of natural gas and electricity services is calculated, and the value of electric boilers that provide the same level of service.

This represents the present value of switching from using a gas boiler to a gas boiler electric.

ÿ =

Finally, the direct costs associated with removing all gas boilers in the selected sectors by 2030 are calculated. This calculation is performed by summing the differences between the costs and benefits of switching from gas boilers to electric boilers, following the demand path shown for each industrial sector.

Additionally, the expected reduction in greenhouse gas (GHG) emissions due to the change from gas boilers to electric boilers is quantified for each of the sectors analyzed.

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Chapter 3. Parameters and results of Scenario 1

The estimates are based on information available from public sources and sectoral studies on final and intermediate natural gas consumption, which are not systematic and have low granularity. Due to these limitations, estimates for the residential sector focus on cooking, while those for the industrial sector focus on boilers for six sectors that account for 20% of total industrial gas consumption (after excluding the coking and refining sectors).

Replacement of assets for stoves, boilers and electricity generation takes place between 2023 and 2030.

3.1. Incremental costs for residential consumers

The calculation of the change in the cost of ownership when switching from a gas stove to an electric induction stove used the following parameters:

- A 12% replacement horizon for gas stoves with induction stoves is considered.
  electricity by 2030 and a social discount rate (
- The top 5 cities were selected to represent consumers
  Colombian residential areas.
- Electricity and natural gas rates correspond to the average for the year 2022
  (Superintendence of Public Utilities, 2022).
- The monthly demand for gas for cooking food is equal to the average consumption of the residential sector (Colombian Energy Balance, 2021) divided by the number of residential subscribers (SUI, 2022).
- The energy efficiency factor of a conventional gas stove is 40% (DOE, 2008).
- The energy efficiency factor of an electric induction stove is 90% (Sadhu et al., 2010).

- The costs of acquiring a new electric induction cooktop are based on quotes
  to large areas (PROURE 2022-2030).

The incremental costs of switching from a gas stove to an electric induction stove in a home, according to its social stratum, are presented in Table 1.

Table 1. Incremental costs of switching from a gas stove to an electric induction stove of a household (COP$), by socioeconomic stratum and in the main cities

	Stratum 1	Stratum 2	Stratum 3	Stratum 4	Stratum 5	Stratum 6
Bogotá	-$2,777,364	-$2,870,050	-$2,777,526	-$3,333,479	-$3,518,851	-$3,518,851
Medellín	-$2,908,790	-$3,034,333	-$3,047,828	-$2,854,113	-$3,913,132	-$3,913,132
Barranquilla	-$3,191,650	-$3,387,908	-$3,628,527	-$3,405,249	-$4,369,196	-$4,369,196
Bucaramanga	-$3,134,724	-$2,584,998	-$3,603,223	-$4,226,881	-$4,590,934	-$4,590,934
Ali	-$2,699,911	-$2,773,234	-$2,515,917	-$3,139,848	-$3,286,493	-$3,286,493

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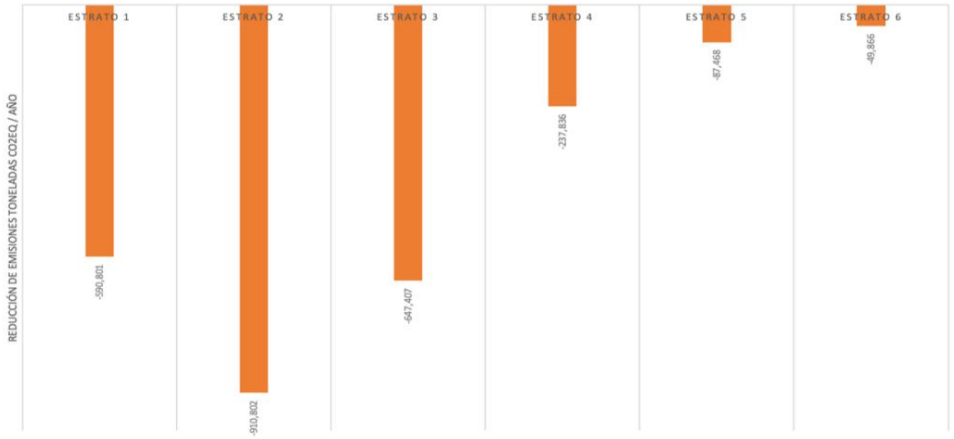
Source: own calculations.

In all socioeconomic strata, the differences between the benefits and the economic costs of switching from a gas stove are negative, meaning there is no economic incentive to switch from a gas stove to an electric induction stove.

The unit costs of gas and electricity are key factors in the cost of switching ownership. For example, in Cali, where the unit cost of gas is higher, the cost of switching ownership is lower. Meanwhile, Bucaramanga has one of the lowest unit costs of gas, meaning that no household has a financial incentive to replace its gas stove.

The annual reduction of greenhouse gases (GHG) due to the change of the stove to gas This is approximately 340 kilograms of CO2eq per household. Figure 12 shows a summary of the greenhouse gas emission reduction for all households in each income bracket after switching to electric induction cooktops, out of the 10.4 million households that use gas for cooking.

Figure 12. Reduction of greenhouse gas emissions when switching from a gas stove to an electric induction stove by stratum (tons CO2eq per year)



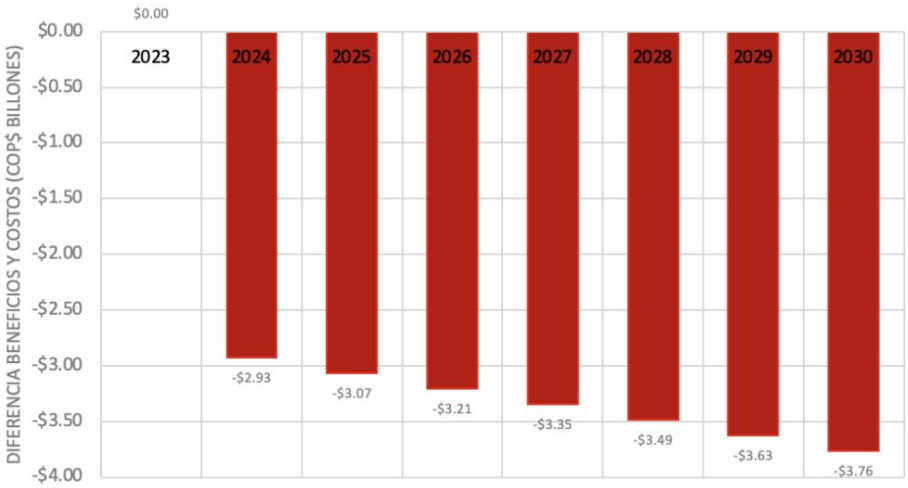
Source: own calculations.

The incremental costs associated with the accelerated phase-out of natural gas cooking in all households by 2030 amount to **COP$18.27 trillion,** including the perpetual value of the energy cost difference, as shown in Figure 13. If all households switched from gas to electricity, the government would cease subsidizing gas tariffs by COP$1.14 trillion annually, but would increase subsidies to electricity tariffs by COP$1.88 trillion. On balance, the increase in government subsidies would amount to **COP$0.74 trillion per year.** The change of 10.4 million

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Households using gas for cooking implies an annual reduction of **2.5 million tons of CO2eq.**

Figure 13. Incremental costs of removing gas stoves and replacing them with induction stoves by 2030 (COP billions)



Source: own calculations.

3.2. Incremental costs for electricity generation

The calculation for replacing gas-fired power generation with wind power generation uses the following parameters:

- A horizon of dismantling and replacement of gas-fired electricity generation with wind power generation by 2030 is considered, and a social discount rate ( ) of 12%.
- The total cost of installing wind power capacity is estimated at USD$1,325 per kW (IRENA, 2023).
- Dollar exchange rate (COP/USD): $4,500.
- Spot price of electricity: marginal costs forecast by UPME (2021).
- Natural gas price (USD/MBTU): UPME forecasts (2021) for reference USD/MBTU December 2020 for Cartagena.
- Heat Rate (MBTU/MWh): 10.16 MBTU / MWh.
- Reliability charge revenue (mm COP): Firm energy obligation * CERE: Actual cost of equivalent energy of reliability change.
- OPEX (COP/kWh): 50 COP/kWh.
- Start-up and shutdown costs (COP): $1,000,000,000.
- Annual number of starts and stops: 15.
- Gas cost (COP/kWh): gas demand (MBTU) * (COP/MBTU) / Total domestic gas price energy sold (kWh).

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• AOM (USD/anual): $ 100,000,000.
• Income tax (mm COP): 35% * EBIT.

As shown in Table 2, natural gas plants have a maximum dispatch factor of 88.10% and wind plants have an average effective capacity factor of 36% (non-dispatchable).

Table 2. Effective capacity and dispatch factor by technology (2015-2023)

	Effective capacity (MW)	Average dispatch factor (%)	Factor of minimum dispatch (%)	Factor of Maximum dispatch (%)
Water	12,549	53.19%	20.12%	70.27%
Gas natural	2,889	33.49%	10.10%	88.10%
Coal	1,664	41.32%	0.00%	95.59%
Wind	18	36%*		
Solar	290	16%*		
Otros3	1,380	27.01%	0.52%	35.79%
Total	18,791	46.83%	34.71%	54.49%

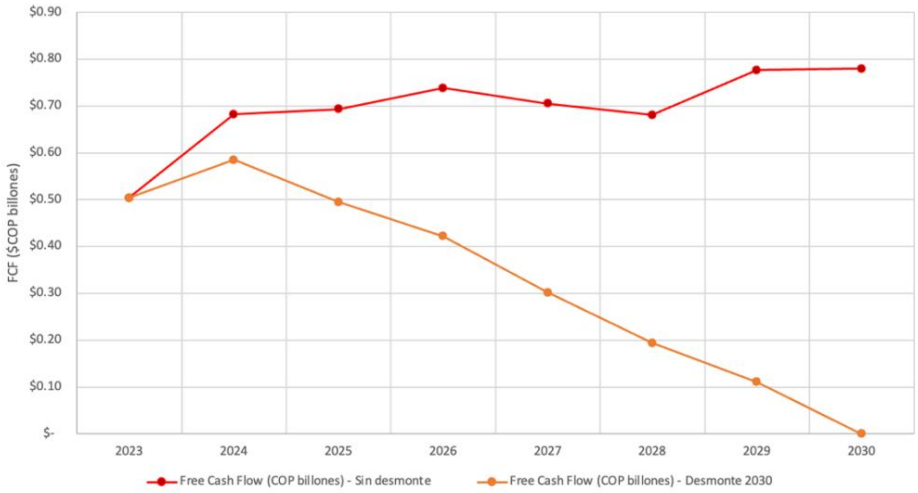
Data source: XM (2023), * IRENA reference values (2023) – Capacity factor (%).

The cost of *replacing* gas-fired thermal power generation capacity with wind power is equivalent to **COP$81.04 trillion.** The cost of stranded assets *for* gas-fired power generation is equal to the net present value (NPV) of free cash flows without decommissioning (red curve), including the perpetuity value, and the net present value (NPV) of free cash flows with decommissioning until 2030 (orange curve), as shown in Figure 14. The difference between these two net present values amounts to **COP$4.32 trillion.**
billions.

[3] Others: ACPM, Bagasse, fuel oil, Jet A-1.

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Chart 14. Estimated free cash flows for thermal – natural gas ($COP billions)



Source: own calculations.

The total cost of decommissioning gas-fired thermal power generation and replacing it with wind power generation amounts to **COP$85.35 billion.** An estimated annual reduction in greenhouse gas emissions is **2.3 million tons of CO2 equivalent,** representing a decrease of approximately 40 grams of CO2 equivalent per kWh generated.

3.3. Costs for the largest industrial consumers

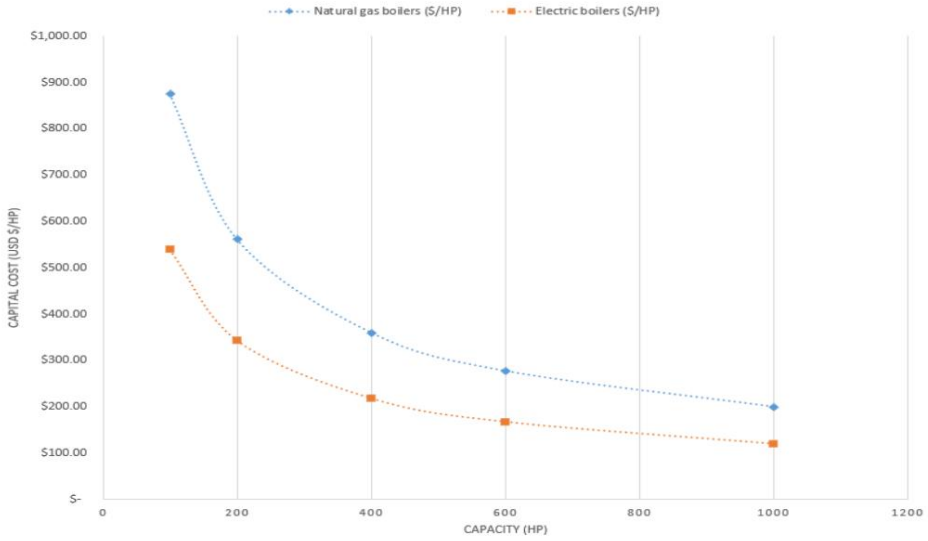
The calculation for replacing gas boilers with electric boilers in the selected industries uses the following parameters:

- An analysis horizon of 2030 and a social discount rate are considered ( ) of 12%.
- The initial inventory and its associated capacity of gas boilers is based on the final report: “Determination of the potential for reducing energy consumption in the manufacturing subsectors ISIC codes 10 to 18 in Colombia” (UPME, 2018).
- The final inventory of gas boilers was carried out by extrapolating the initial inventory with the current consumption of each manufacturing subsector.
- The efficiency of conventional gas boilers is assumed to equal 75% (Wouter et al., 2010).
- The efficiency of conventional electric boilers is assumed to be 99% (Schoeneberger et al., 2019).
- Exchange rate: dollar (COP/USD): $4,500.
- Electricity and natural gas rates correspond to the average for the year 2022 (Superintendence of Public Utilities, 2022).

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• The costs of acquiring a new electric boiler (NREL, 2017) and commercial quotes for Viessmann Vitomax 100-1 boiler, Bosch 1000-1000 boiler, and Burnham 1000HP boiler.

Chart 15. Installation costs of gas boilers and electric boilers (USD$ per HP)

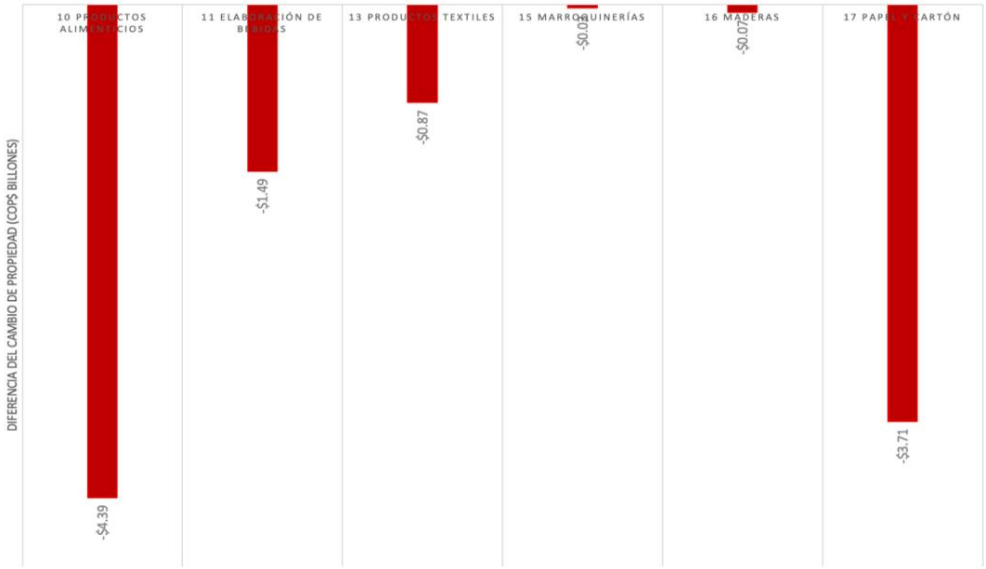


Data source: NREL (2017) and commercial quotations (Viessmann Vitomax 100-1 boiler, Bosch 1000-1000 boiler, and Burnham 1000 HP boiler)

When applying the cost calculation methodology to the six industry sectors, the cost difference between gas boilers and electric boilers is negative in all subsectors, meaning that no sector has economic incentives to switch to gas boilers (see Graph 16). The present value of the costs of switching from gas to electric boilers in the six sectors by 2030 amounts to **COP$10.57 trillion,** an amount that includes the perpetuity value of the energy cost difference.

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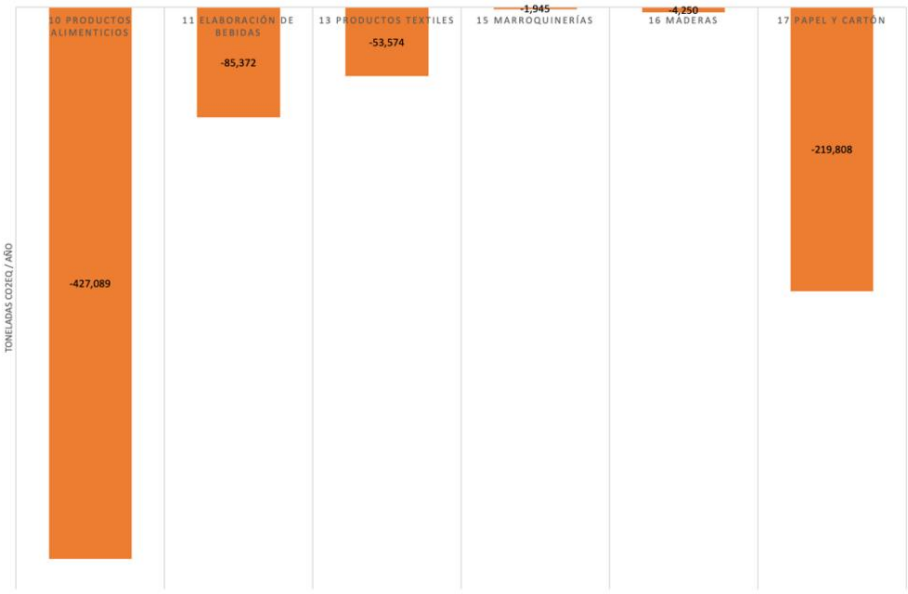
Chart 16. Difference in total cost of ownership for each subsector (COP$ billions)



Source: own calculations.

The annual reductions in greenhouse gas emissions for these six sectors would be equivalent to **0.8 million tons of CO2eq.** The expected emissions reduction for
Each sector is shown in Graph 17.

Figure 17. Estimated reduction of greenhouse gases by subsector (tons CO2eq per year)



Source: own calculations.

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Chapter 4. Methodology for estimating incremental costs of Scenario 2

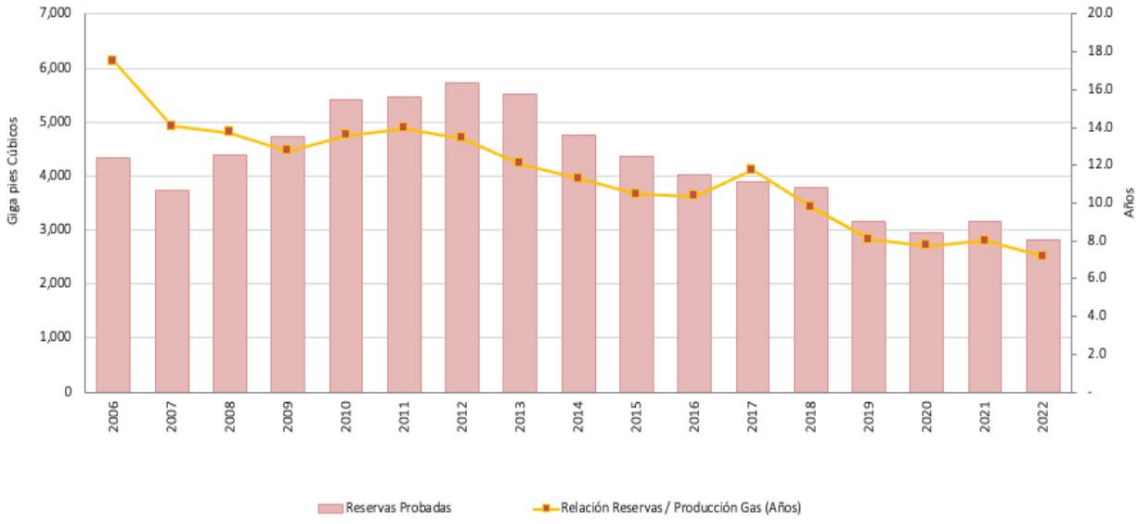
Scenario 2 consists of (i) halting new exploration; (ii) expanding import infrastructure; (iii) maintaining gas-fired power generation capacity; and (iv) maintaining natural gas consumption in industry and households. In cases (ii) and (iii), the reduction in domestic consumption is achieved with increasing proportions of imported gas.

4.1. Context for a scenario of imports and international prices

In 2022, average marketed natural gas production was 392 billion cubic feet (Bcf), and proven natural gas reserves amounted to 2,817 Bcf (ANH, 2023). These reserves are sufficient to cover approximately 7.2 years at the current production rate (Figure 18).

To increase the supply of domestic gas, proven reserves can be added through exploratory well discoveries, enhanced recovery techniques from mature wells, economic considerations, reclassifications, or technical reviews. For example, in 2022, 45 billion cubic feet (Bcf) were added to proven reserves due to these factors, representing 11.48% of marketed production. Therefore, if the addition of proven reserves is less than production, gas production will be depleted.
natural becomes inevitable.

Figure 18. Proven reserves (Gpc) and the reserves-to-production ratio (years)



Source: ANH (2023).

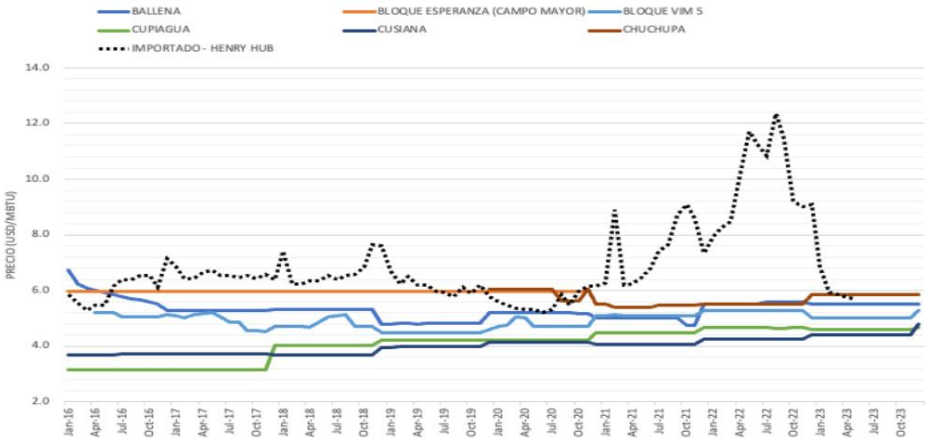
In 2016, Colombia joined the international Liquefied Natural Gas (LNG) market with the commissioning of a terminal called SPEC LNG in Cartagena, which allows for the reception, storage, and regasification of imported liquefied gas.

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It supports some thermal power generation plants, which provides reliability to the electrical power system in Colombia.

On the other hand, in recent years, the prices of contracts guaranteeing gas firmness in the Cusiana and Cupiagua fields have shown a similar upward trend, as seen in Figure 19. However, these differ from the price trends in the La Guajira fields (Ballena and Chuchupa) and the Esperanza and VIM 5 block fields. Finally, it can be observed that most of the time the price of imported gas is higher than the price of domestic gas. 4

Chart 19. Average price of current contracts guaranteeing firmness by supply source - Primary Market (2016-2023)



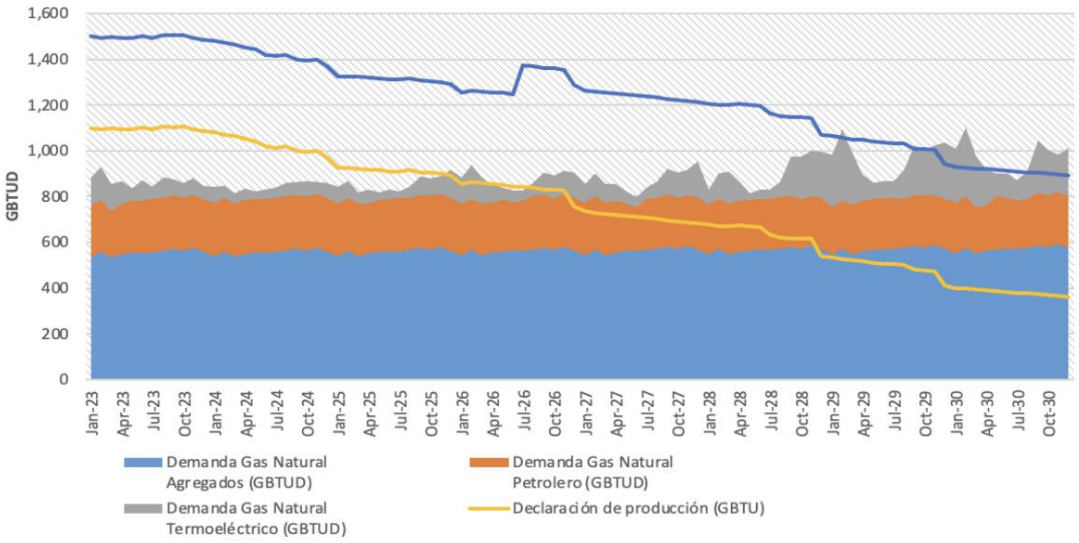
Source: Colombian Mercantile Exchange (2023).

When taking into account the UPME's (2022) average scenarios for aggregate natural gas demand (residential, industrial, commercial, vehicular, among others), the oil sector and the thermoelectric sector, and the declaration of production potential, we observe a deficit in domestic gas supply from 2026 onwards, which will have to be supplied with imported gas (Graph 20).

[4] Imported gas prices are the result of aggregating the international price (Henry Hub), the liquefaction cost, the shipping cost, the regasification cost, and marketing.

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Figure 20. Natural gas demand - UPME medium scenario (2022)[5] - versus production declaration (GBTUD) plus LNG import capacity



Source: UPME (2022) & Ministry of Mines and Energy (2023).

4.2. Methodology for estimating the incremental cost of the system when performing gas imports without reserve addition

To calculate the additional cost of maintaining gas demand without adding new gas reserves, the following steps must be followed:

(i) A nodal model is built and calibrated to simulate the natural gas transportation system in Colombia, based on the sections established in the transportation tariff resolutions of the Energy and Gas Regulatory Commission (CREG), as shown in Graph 21. This model includes a total of 29 nodes representing producers, LNG import port, consumers and compression stations.

Without the addition of new proven reserves, primarily through exploration, the natural gas deficit must be covered by importing liquefied natural gas (LNG). Importing natural gas involves gas liquefaction, maritime transport, and regasification before it can be incorporated into the system. Imported natural gas has direct effects on gas service tariffs, both in terms of the cost per molecule (G) and transportation (T).

[5] The UPME's medium scenario (2022) includes the annual demand for the thermoelectric sector with an average growth of 15.40%.

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Figure 21. Gas transport network structure for modeling



Source: own elaboration.

The nodal model of the natural gas transportation system has the objective function of minimizing the total cost of natural gas supply (ÿ). This total cost is composed of the wellhead gas supply cost, the total transportation cost, and the rationing cost for the national gas system.

$$\min \ \ddot{y}t = \ddot{y}_{=1} \quad +\ddot{y}\ddot{y}_{=1\ =1} \quad +\ddot{y}_{=1}(\ddot{y})$$

Subject to the following restrictions:

$$\ddot{y}_{=1} \quad + = + \ddot{y}_{=1} \quad .\ddot{y}$$

$$0 \ddot{y} \ddot{y} \quad .\ddot{y}$$

$$0 \ddot{y} \ddot{y} ,\ddot{y}$$

$$0 \ddot{y} \ddot{y} \quad .\ddot{y} \ddot{y}$$

Where:

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= wellhead price at the node = quantity of

gas produced at the node = transportation rate

from node to node

= volume transported from node to node

= nationally weighted demand rationing cost per unit volume, defined as the measure of economic or social damage caused by the interruption of natural gas service = demand required at the node = volume supplied from the node's demand

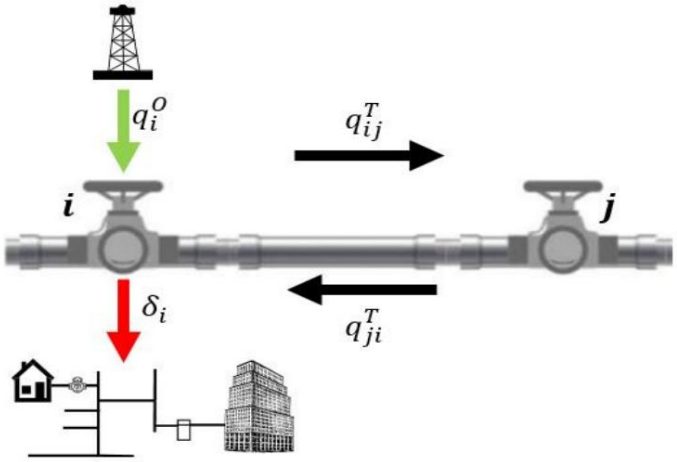
= maximum production capacity of the node = maximum

transport capacity between node to node

The first constraint is a balance equation. Therefore, as shown in Figure 22, the sum of the gas flow and the flow arriving at the node from any node (ÿ

=1), along with the production of the node) must be equal to the actual demand) and what (supplied to the node (of comes out of the node for any of the j nodes in the network gas (ÿ =1).

Figure 22. Natural gas nodal network balance equation



Source: own elaboration

The second constraint of the optimization model is associated with the security of supply of the node, where the demand actually supplied to node *i* ( ) must be less than or equal to the demand of node i ( ) and greater than zero. When this constraint is not active (=) in the optimization model, rationing occurs at the node, which is equivalent to the difference between demand and supply (ÿ).

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Finally, the third and fourth optimization model constraints are the upper and lower bounds for the production of a node ( ),) and pipeline transport (respectively.

Based on the optimal solutions of the natural gas system, the following are calculated steps.

ii) If new exploration is not stopped, new proven reserves may be incorporated.
Under this scenario, the present value of the cost of natural gas is estimated.
(with from), to a horizon of T years:

$$\text{with} \quad \text{from} \quad = \ddot{y}_{=1} \frac{\ddot{y}_{=1}}{(1+\quad)}$$

iii) If new exploration is stopped, no significant proven reserves will be added.
Under this scenario, the present value of the cost of natural gas is estimated.
( ), given sin the current from proven reserves in Colombia, in a
T-year horizon:

$$\text{sin} \quad \text{from} \quad = \ddot{y}_{=1} \frac{\ddot{y}_{=1}}{(1+\quad)}$$

iv) Finally, the difference between the present value of the cost of natural gas is calculated if not the new exploration is halted (with from) the present value of the cost of the gas natural if new exploration is stopped (represents the sin from This difference increase in the cost of gas due to gas imports).

$$\ddot{y} \quad = \quad \text{with} \quad \text{from} \quad , \quad \text{sin} \quad \text{from}$$

v) If the new exploration is not stopped, the present value of the transportation cost is estimated.
natural gas (with from), to a horizon of T years:

$$\text{with} \quad \text{from} \quad = \ddot{y}_{=1} \frac{\ddot{y}\,\ddot{y}_{=1\ =1}}{(1+\quad)}$$

vi) If the new exploration is stopped, the present value of the transport cost is estimated over a natural gas (sin from horizon of T years.

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$$\text{sin} \quad \text{from} \quad = \ddot{y}_{=1} \frac{\ddot{y}\ \ddot{y}_{-1\ =1}}{(1 + \qquad)}$$

vii) The difference between the present value of the natural gas transportation cost is calculated if The new exploration does not stop (with from) and if the new one stops exploration sin from). This difference represents the increased cost of (gas transport due to gas imports).

$$\ddot{y} \qquad = \text{with} \quad \text{from} \quad , \quad \text{sin} \quad \text{from}$$

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Chapter 5. Parameters and results of Scenario 2

As in Chapter 3, the estimates were made using information available from public sources and sectoral studies on final and intermediate consumption of natural gas, which are not systematic and have low granularity.

5.1. Incremental costs for residential consumers

The assumptions for the nodal simulation model of the National Transportation System (NTS):

- The analysis horizon is 2023 to 2030.
- The annual discount rate (• It is) is 12.00%.

assumed that the calorific values of all gas sources are equivalent to 1000 British Thermal Unit (BTU) per cubic foot (CF) of natural gas.

- Imported gas prices result from the sum of the international price (Henry Hub), the liquefaction cost, the shipping cost, the regasification cost and the marketing cost.

- International price projections are provided by the EIA (2023). Costs are adjusted using the annual average of the Producer Price Index (PPI) - Industrial Gas Manufacturing (FRED, 2023), and shipping costs are adjusted using the annual average of the Producer Price Index (PPI) (FRED, 2023), as presented in the Table 3.

Table 3. Projections of the price of natural gas imported by Cartagena and Buenaventura.
Source: Prepared by the author based on data from the EIA (2023), UPME (2022) and FRED (2023)

	2023	2024	2025	2026	2027	2028	2029	2030
Natural gas spot price at Henry Hub (THIS)	3.10	2.95	3.10	3.29	3.44	3.68	3.91	4.08
Liquefaction cost6	2.19	2.26	2.34	2.42	2.50	2.58	2.67	2.76
Regasification cost (UPME)	0.79	0.81	0.84	0.87	0.90	0.93	0.96	0.99
Marketing cost (UPME)	0.18	0.19	0.20	0.20	0.21	0.22	0.22	0.23
Cost of maritime transport - Cartagena (UPME)	0.40	0.40	0.41	0.41	0.42	0.42	0.43	0.43
Natural gas price - Cartagena	6.66	6.62	6.88	7.19	7.46	7.83	8.20	8.50

- Since historical series or price projections are not available for each oil/gas field, three wellhead prices are proposed: Cusiana, Ballena and others

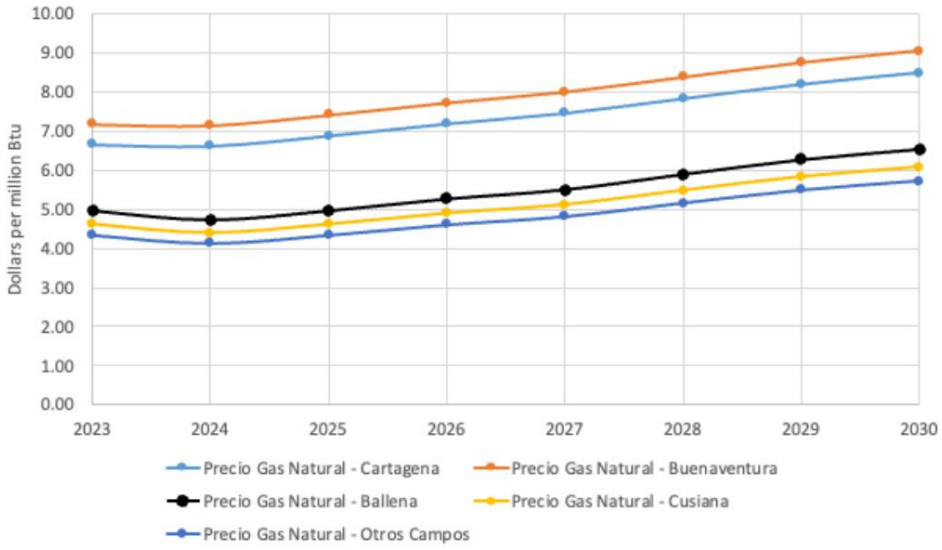
[6] The liquefaction cost was calculated as the average of the differences since January 2016 between the Price of Liquefied US Natural Gas Exports (Dollars per Thousand Cubic Feet) and the Price of US Natural Gas Pipeline Exports (Dollars per Thousand Cubic Feet) reported by the US Energy Information Administration https://www.eia.gov

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fields. Prices are calculated as the 2022 weighted averages for each firm contract type (BMC, 2023). They are projected using the percentage changes in the international Henry Hub price projections (EIA, 2023). Wellhead and imported price projections can be seen in the

Graphic 23.

Chart 23. Projections of wellhead and imported prices



Source: own elaboration based on data from EIA (2023), UPME (2022) and FRED (2023).

- Transportation rates are based on CREG resolutions and are adjusted according to the average of the last 10 years of the producer price index in dollars (PPI)
  (FRED, 2023).
- The cost of natural gas rationing is defined as the measure of the economic or social damage caused by the interruption of natural gas service (UPME, 2019). Its value is assumed to be 32 USD/kPC (Econometría SA, 2015) and is adjusted with the producer price index in dollars (PPI) (FRED, 2023).

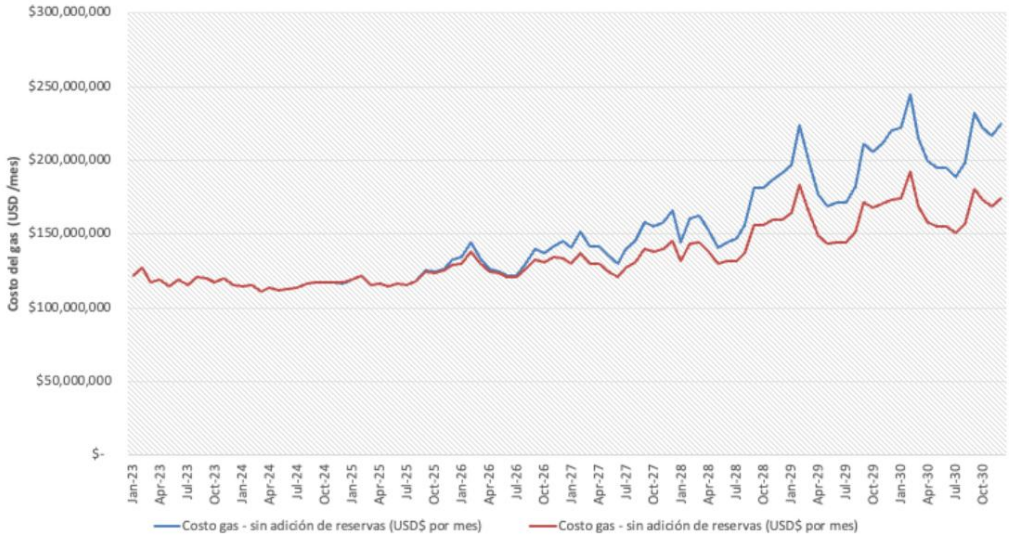
- The aggregate natural gas demand series for each node are used, which are associated with the medium scenario developed by UPME. Aggregate natural gas demand comprises the sum of demand from the residential, commercial, industrial, petrochemical, compressor, CNG (compressed natural gas) transportation, oil, and thermoelectric sectors. On the other hand, the domestic natural gas supply series for each node are obtained by aggregating the production declarations for each oil/gas field. In addition to domestic supply, there is the international supply of natural gas, which can be liquefied natural gas (LNG) via the SPEC LNG regasification plant in Cartagena, with an initial capacity of 400 MMcf/d and an expansion to 530 MMcf/d by mid-2026. Currently, the available capacity

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Importing liquefied natural gas by SPEC LNG is subject to the use of thermal generation.

Before 2026, no gas imports will be required to meet domestic demand, according to UPME's medium scenario. After 2026, without the addition of new gas reserves, the monthly cost of gas is higher compared to the scenario that includes reserve exploration, as shown in Figure 24.

Figure 24. Cost of the gas molecule for the entire system (USD$ per month)



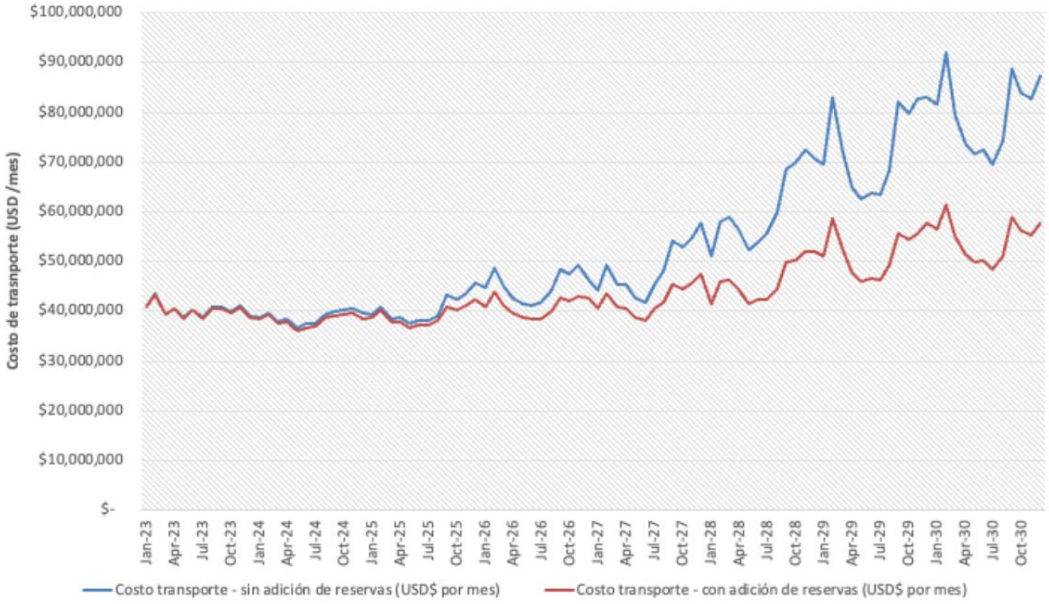
Source: own elaboration

The present value of the cost of increasingly importing gas to compensate for the slowdown in gas exploration is **COP$3.14 trillion.** By the end of the decade, the portion of the tariff associated with the value of the gas molecule could increase **by 28.71%** compared to the scenario that allows exploration.

Without the incorporation of new gas reserves, the monthly cost of gas *transportation* due to the reconfiguration of flows in the transportation system is higher compared to the scenario that includes reserve exploration, as shown in Figure 25. This is explained by the increased distance between the import infrastructure and the main gas consumption centers, compared to the domestic gas supply sources.

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Figure 25. Gas transportation cost for the entire system (USD$ per month)



Source: own elaboration

The present value of the cost of increasing imports to compensate for the slowdown in gas exploration is **COP$3.14 trillion.** The difference in the present values of gas transportation costs in 2030, between the scenario with and without exploration, is approximately **COP$2.00 trillion.**

Without the addition of new gas reserves, the increase in the present value of the cost to supply gas to Colombia until 2030 through increasing imports amounts to **COP$ 5.14 trillion.**

By the end of the decade, the natural gas service tariff will experience an additional increase of 28.71% in gas costs and 50.95% in transportation costs, compared to the scenario with new reserves. This could imply **an additional increase of more than 25%** in the final tariff.

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Chapter 6. Core Messages

The central messages of this article are as follows:

• No single energy source dominates the others in terms of cost, service continuity, emissions, and volumetric and area density. Natural gas provides firm energy for electricity generation, dense energy for industry and residential consumption at lower costs than liquid fuels, and emits fewer greenhouse gases per unit of energy consumed.

The energy crisis in Western Europe, triggered by Russia's natural gas cutoffs, highlights the precariousness of attempting to electrify the entire energy supply with unconventional renewable energy sources and to electrify all final consumption in the short term. Countries are beginning to understand that they must build energy supply portfolios that balance security, diversification, and the reduction of greenhouse gas emissions.

The costs of either scenario overlap with persistent inflationary trends in energy prices observed internationally. In scenario 1, switching from gas cooking to electric induction cooktops in 10.4 million households, replacing gas-fired power generation with wind power, and replacing gas boilers with electric boilers in some productive sectors would reduce CO2 equivalent emissions by 5.6 million tons annually. This would represent 3.2% of the government's commitment to reducing greenhouse gas (GHG) emissions by 2030, at a cost of COP$114.19 trillion. Scenario 2, in turn, highlights the importance of not neglecting domestic natural gas production.

The enormous financial effort with such a low impact on the government's commitment is not sustainable for the Colombian economy. Both scenarios demonstrate that natural gas must play a more significant role in the energy portfolio in an international environment where energy security is regaining prominence, and given the threats of electricity generation shortages due to climate shocks such as

The El Niño phenomenon.

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